

FACULTY ELECTRICAL ENGINEERING, MATHEMATICS
AND COMPUTER SCIENCE

Department of Microelectronics

[EE4520] Analogue CMOS Design I

Homework II [Fully Differential Folded-Cascode Amplifier]

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1 Deliverable 1 [Table of Results]

Table 1: Table of Results

Item	Design Item	Unit	Spec.	Achieved
1	Design Number	70		
2	SNR	[dB]	83.6	83.6
3	A_{settle}	[dB]	54	54
4	T_{settle}	[μ s]	1.11	1.11
5	BW_{cl} from open-loop AC simulations	[MHz]	–	0.7029
6	BW_{cl} from closed-loop AC simulations	[MHz]	–	0.70244
7	Time-constant τ_{cl} from closed-loop AC simulation	[μ s]	–	0.2266
8	T_{40dB} (time to reach an accuracy of 40dB)	[μ s]	–	0.6722
9	$T_{48.69dB}$ (time to reach an accuracy of 48.69dB)	[μ s]	–	0.9299
10	Time constant τ_{cl} from transient simulation	[μ s]	–	0.2577
11	Power dissipation	[μ W]	Lowest Possible	357.336
12	$I(V_{dd})$	[μ A]	Lowest Possible	198.52
13	Total integrated noise	[μV_{rms}]	–	56.06
14	Input step voltage	[mV]	150	150
15	Output step voltage	[V]	1.2	1.2
16	C_{in}	[pF]	–	26.821
17	C_{fb}	[pF]	–	3.352625
18	C_{load}	[pF]	–	26.821
19	C_{CM}	[pF]	–	3.352625
20	R_{big}	[G Ω]	Big	1000
21	C_{big}	[uF]	Big	1
22	Bonus points FoM_{dB}		0.05	
23	$FoM_{lin} = 2\pi * Power * T_{8.6dB} / SNR_{lin}^2$	[J]	$\leq 3.98e - 18$	$2.5255 * 10^{-18}$
24	$FoM_{dB} = -10 \log_{10}(FoM_{lin})$	[dB]	≥ 174.0	175.98

2 Deliverable 2 [Schematic with all node voltages]

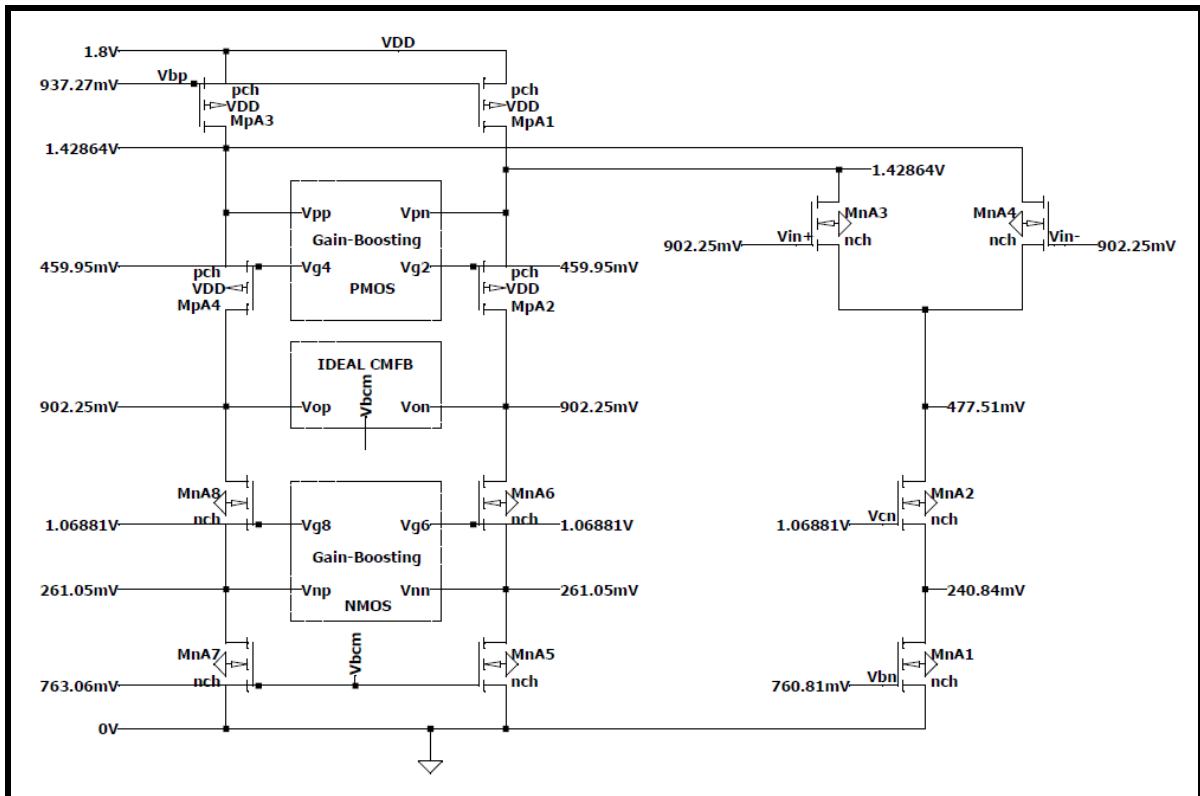


Figure 2.1: Schematic with all node voltages.

3 Deliverable 3 [Schematic with all branch currents]

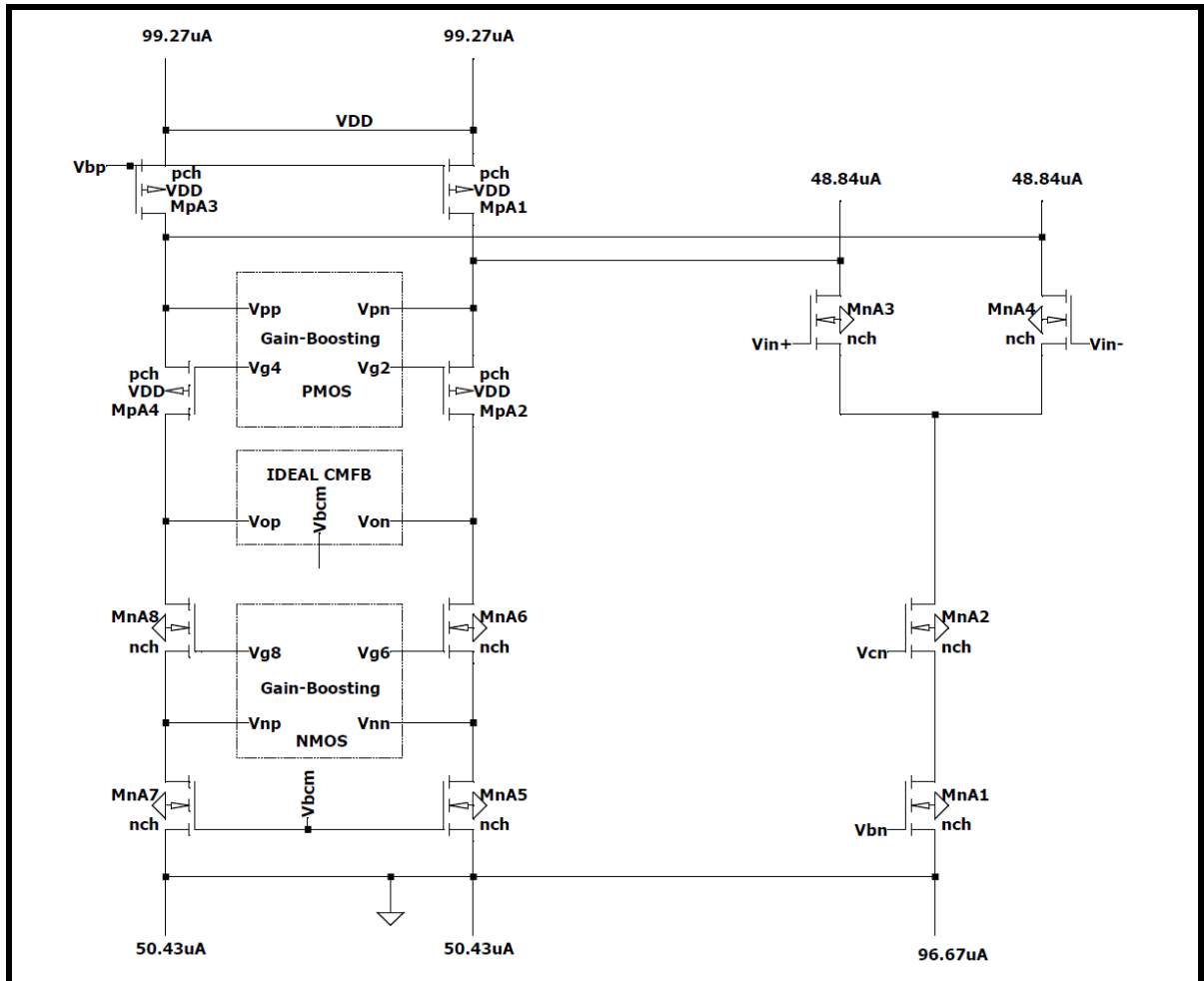


Figure 3.1: Schematic with all branch currents.

The currents are captured from the operating point simulation where the differential input is 0 V. The value in the table have a deviation 0.01-0.02uA since it was captured from the active simulations of noise and transient analysis.

4 Deliverable 4 [Bode-diagram of A and Aβ with Gain (dB) and Phase]

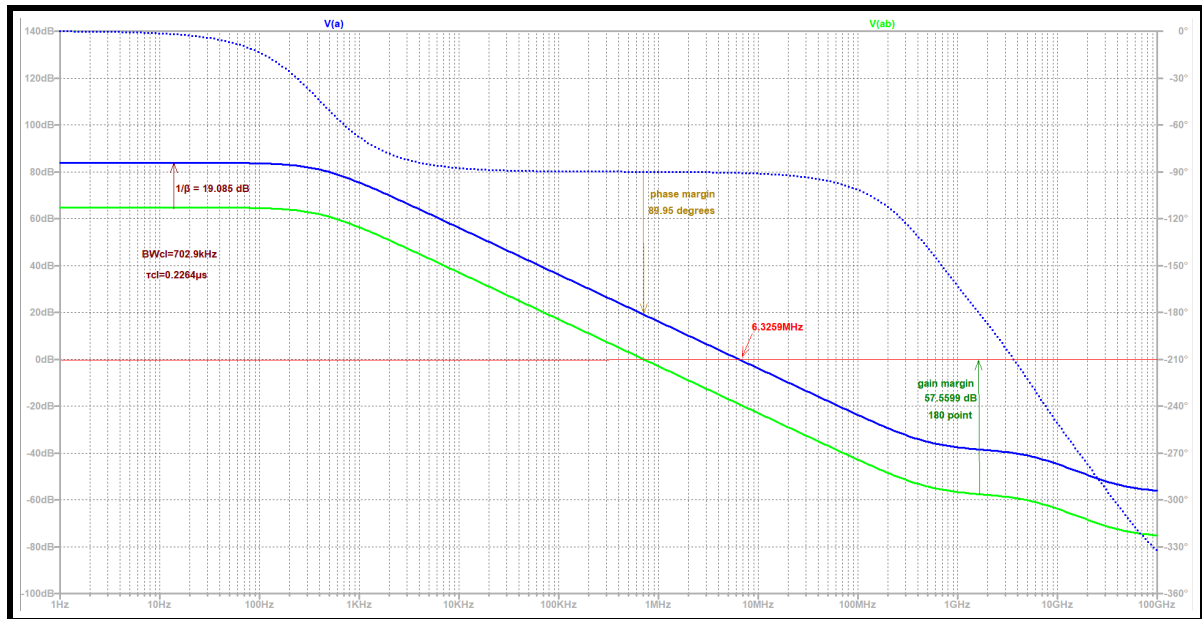


Figure 4.1: Open-loop gain ($A(j\omega)$) and loop gain ($A\beta(j\omega)$).

- gain margin: 57.5599dB
- phase margin: 89.95 degrees
- unity gain frequency: 6.3259MHz
- $1/\beta$: 19.085dB

The loop gain and the open-loop gain have a distance of $1/\beta$ vertically. I know that the product of the unity gain frequency of the open-loop gain and the β gives me the bandwidth of the closed loop-gain, so:

$$BW_{cl} = \beta \omega_u \Rightarrow$$

$$BW_{cl} = 0.7029 \text{ MHz}$$

The time constant of the closed-loop gain is:

$$\tau_{cl} = \frac{1}{2\pi BW_{cl}} \Rightarrow$$

$$\tau_{cl} = 0.2264 \mu\text{s}$$

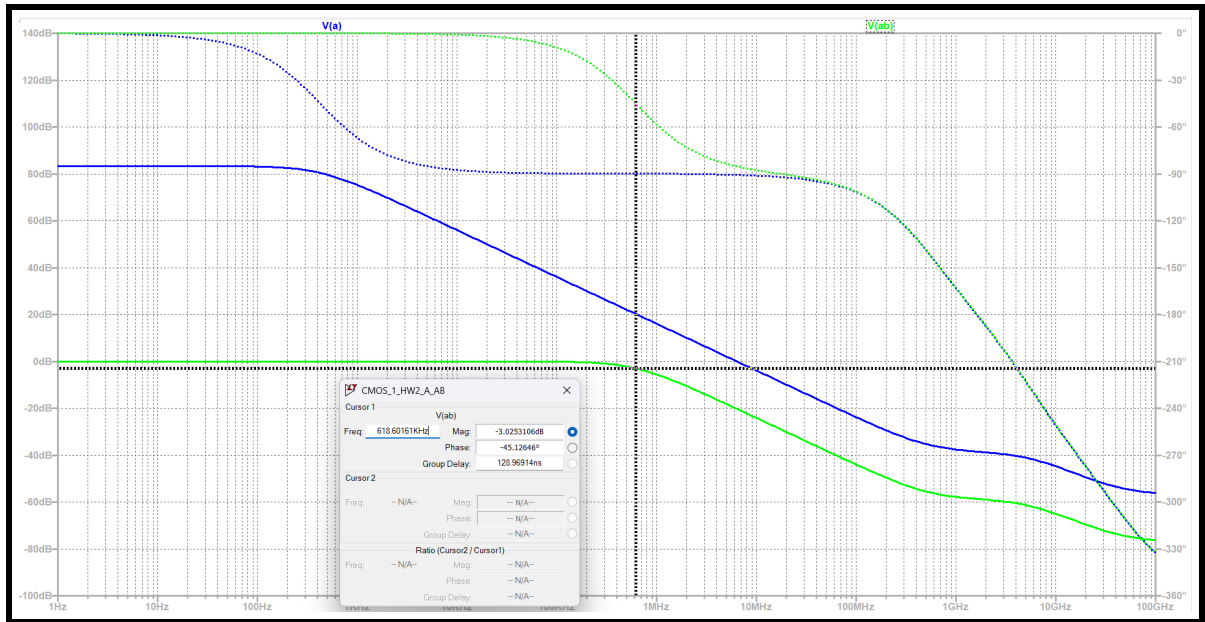


Figure 4.2: Open-loop gain ($A(j\omega)$) and loop gain ($A\beta(j\omega)$) with the β -network loaded.

By examining the figure (4.2), we derive that the open-loop gain is not affected by the extra load, while the bandwidth of the loop-gain has been altered. Specifically, it has been decreased due to the extra parasitic capacitance that the loaded amplifier introduces. Lastly, at low frequencies the loop gain stays at zero dB since the feedback path acts as a buffer.

5 Deliverable 5 [Bode-diagram of closed-loop Gain with Gain (dB) and Phase]

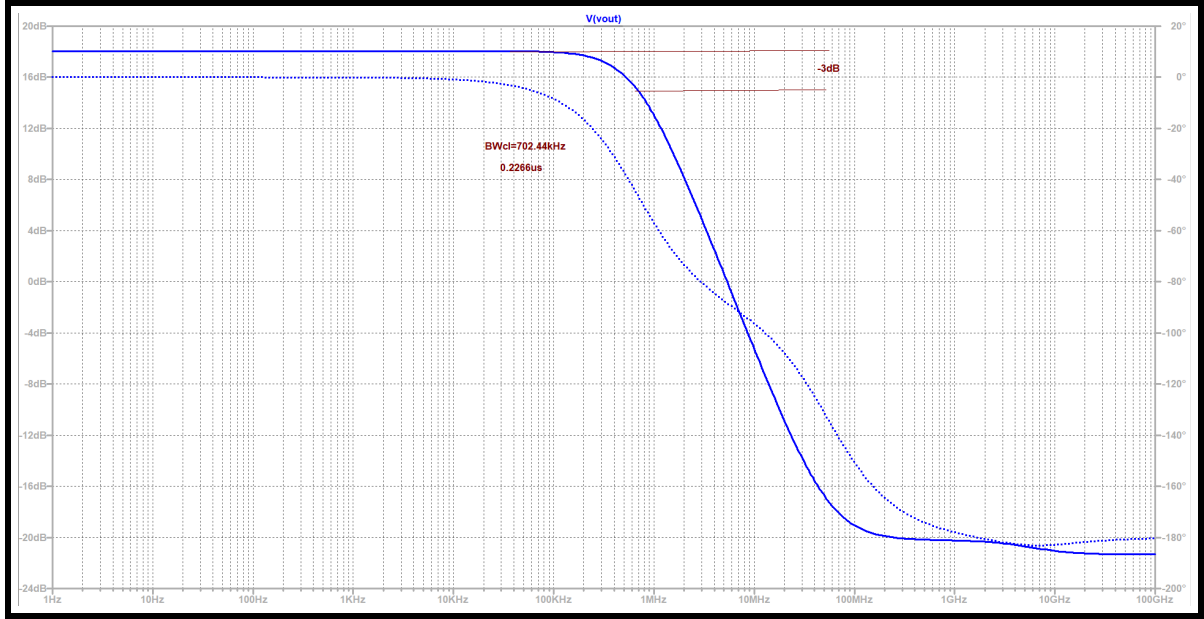


Figure 5.1: Closed-loop gain.

According to the graph the closed-loop bandwidth is:

$$BW_{cl} = 702.44kHz$$

The time constant is:

$$\tau_{cl} = \frac{1}{2\pi BW_{cl}} \Rightarrow$$

$$\tau_{cl} = 0.2266\mu s$$

We can observe that the time constant τ_{cl} from the close-loop AC simulation is close to that calculated from the open-loop AC simulation. I believe that the small difference exist due to the fact these measurements have been carried out with a cursor rather than a function (.meas) as it was used in the settling simulation.

6 Deliverable 6 [Settling accuracy versus time]

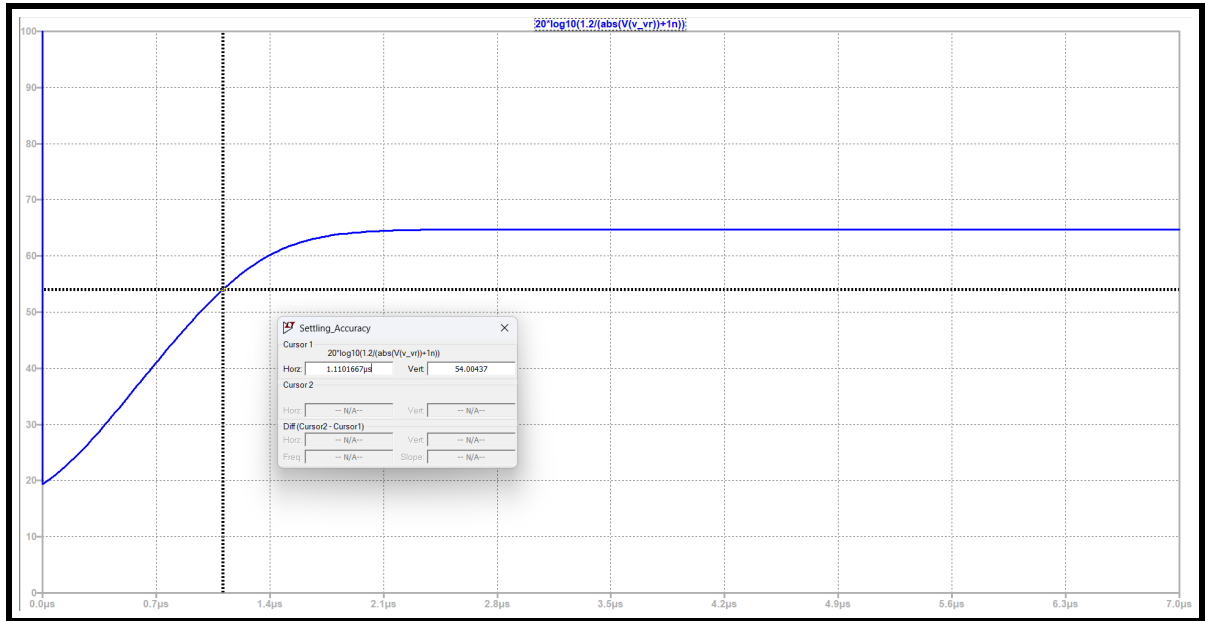


Figure 6.1: Settling accuracy versus time.

The expression that we follow for the transient analysis is:

$$Accuracy_{dB} = 20 * \log_{10}\left(\frac{1.2V}{|V_{virtualground}| + 1n}\right) \quad (6.1)$$

7 Deliverable 7 [Settling accuracy versus time with estimated time-constant τ_{cl}]

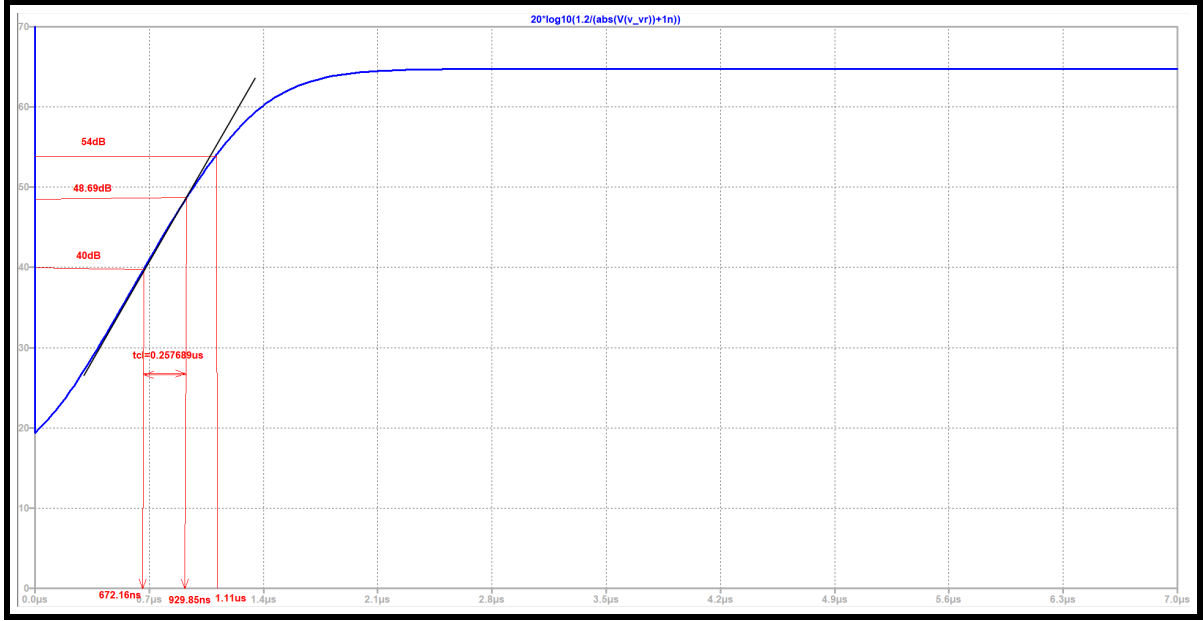


Figure 7.1: Settling accuracy versus time with graphics.

- Time to reach 40dB: $T_{40dB} = 672.16ns$
- Time to reach 48.69dB: $T_{48.69dB} = 929.85ns$
- Time to settle: 1.11μs

The closed-loop settling time constant is found by subtracting those values:

$$\tau_{cl} = T_{48.69dB} - T_{40dB} = 929.85ns - 672.16ns \Rightarrow$$

$$\tau_{cl} = 0.257689\mu s$$

We can clearly see that in this case the closed-loop time constant is larger than those calculated on the open-loop and closed-loop simulations. The reason is that in the transient analysis, the nonlinear effects are also included in contrast to the previous simulation. Moreover, the Bandwidth is smaller than the previous simulations since it is inversely proportional with the closed-loop time constant.

8 Deliverable 8 [Plot of output noise power density versus frequency]

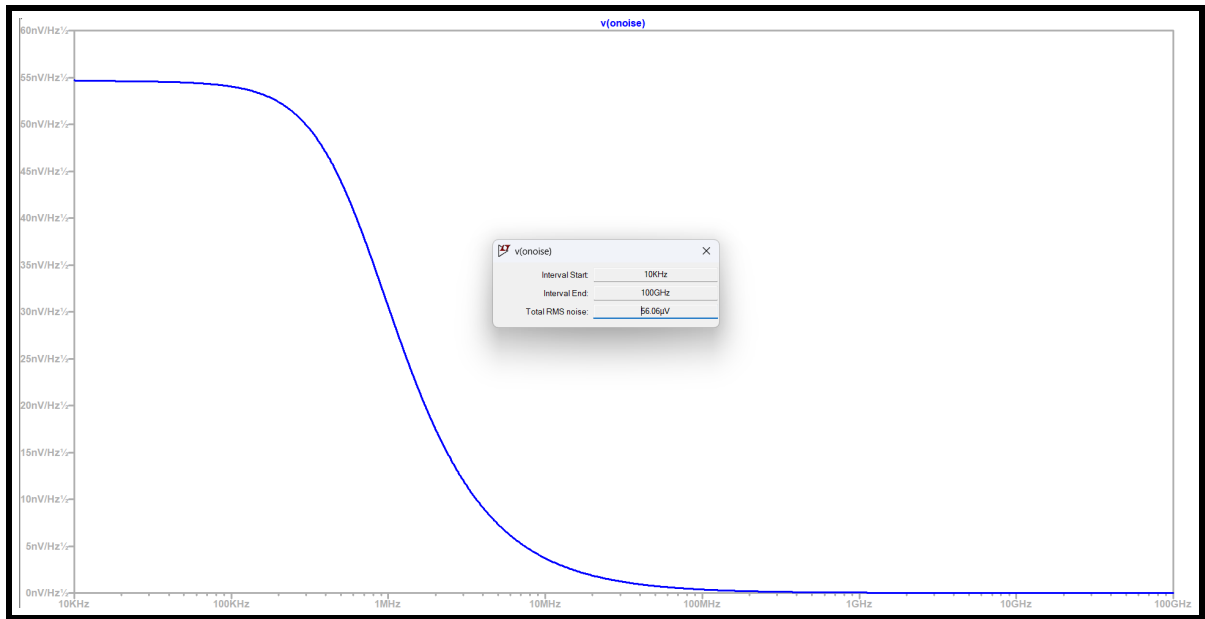


Figure 8.1: Output noise spectral density and integrated noise.

The RMS value of the signal is:

$$V_{sigRMS} = 0.8485V$$

The RMS value of the noise is:

$$V_{noiseRMS} = 56.06\mu V$$

The resulted SNR is:

$$SNR = 20\log\left(\frac{V_{sigRMS}}{V_{noiseRMS}}\right) = 20\log\left(\frac{0.8485 * 10^6}{56.06}\right) \Rightarrow$$

$$SNR = 83.5999dB$$

9 Deliverable 9 [Design Process]

9.1 Parameters

Before starting to design, it is essential to describe what effect have the parameters on our design:

- **scaleA**: It controls the width of the devices on our telescopic folded cascode. By increasing this parameter the current that flows through the transistor rises, increasing their g_m and as a result increasing the bandwidth. This makes our circuit faster (settling), but also it affects our SNR, since by increasing the bandwidth we allow more noise to enter our signal. Moreover, more noise is produced by the transistors that do not contribute to the amplification (current transistors).
- **C_L** : Similarly with the **scaleA**, it affects the bandwidth since the first pole in the most systems is dominated by it. Decreasing its value increases the bandwidth and as a result it affects the settling and the SNR.
- **Ibmain**: This is the current source at the biasing. By increasing its value, the V_{gt} 's of the transistors rise. This parameter in general increases the SNR since the transistors that it affects are the current source transistors. These transistors do not amplify any signal, but they just provide current. Their g_m should stay low in order to minimize their noise contribution. By biasing them in a higher V_{gt} the g_m efficiency drops, meaning that with the same current they generate less g_m .
- **scale of Mn3 and Mn4**: This parameter is important for several reasons. The first one is that it helps us bias the input transistor in weak inversion. Let's assume that we don't change the current that our circuit draws, and we increase the widths of the input transistors. The current wants to become bigger, but it can't, thus the voltage of their source node is increased. This way the V_{gt} drops, the efficiency of the g_m increases and as a result the SNR is increasing because the amplification of the signal is stronger than that of the noise. Moreover, they contribute to the settling because they generate more g_m for the same current, meaning more bandwidth. Moreover, they are really important for the headroom of the tail transistors. If any transistor goes in to triode we can adjust the headroom through them.
- **Ibpcas**: this is the current source that controls how much current will be drawn from the PMOS transistors in the biasing. In general, lowering the current of this source contribute to bias the PMOS current sources of the amplifier in a lower V_{gt} , which is what we want to have a higher SNR. The short reason is; we decrease the current flowing through the Mp3 so its gate increases, the current in Mp2 is constant due to Ibmain meaning that the source of the Mp2 will increase and as a result the gate voltage of the Mp1 will decrease since it wants to keep its current stable.
- **Ibncas**: Lowering this current source will increase the V_{gt} of the sunk NMOS transistors following a similar logic as Ibpcas.
- **Mp, Mn**: These are parameters that affect the biasing and unlike the two

current source that were described before they affect the amplifier in two ways; both through the biasing of the gates but also through the M_p/M_n ratio. Since the effect of them start to become more complicated I used them at the end of the design, for fine-tuning and to drag some transistors out of triode.

9.2 Design Steps

Firstly, I used exactly the values indicated by the HW2 and the previous these elaborated in HW1, and I performed a noise simulation with $scaleA=1$ and a random value of C_L in the range of picofarads. Afterwards, I started applying the techniques described above to reduce the noise.

I started by increasing the I_{bmain} and the widths of M_n3 and M_n4 , while constantly checking the operating conditions of my transistors. Afterwards, I lower my current sources of I_{bpcas} and I_{bncas} , until I got to a point where I was close to the saturation condition of the transistors.

Afterwards, I increased the C_L to lower the SNR and run a settling analysis. At this point, even though I increased the C_L , the settling was in a good state, so I decrease the $scaleA$.

Lastly, as I mentioned, I tried to optimise my design to lower the power consumption, mainly by trying to decrease the $scaleA$. A solution was to increase greatly the widths of the input transistors. However, after a certain point (at 1500 approximately), I stopped progressing in my SNR but only on settling. The increase of the widths started to lower my SNR since the input transistors introduced large parasitic capacitances. However, the positive effect on settling was greater, so I compensated by increasing my C_L a little every time.

My parameters were:

- $C_L = 26.821P$
- $I_{bmain} = 800u$
- $M_n=15.17$
- $M_p=36$
- $scaleA=0.4707$
- $M_n34=2500$
- $I_{bncas}=3.86$
- $I_{bpcas}=4.66$